

DGGR Final Presentation Notes

Simple glanceable notes for the 10-slide main deck.

Slide 1: Deep Generative Genre Remastering

- Open by contrasting coat-of-paint style transfer with controlled genre remastering.
- Frame the project as: separate content and style, build a target space, generate toward that target, then make it survive long-form playback.
- Point to the pipeline strip and the content / realism / style tradeoff.
- Keep this slide conceptual and do not play audio yet.
- End by saying the first two labs matter because without trustworthy control, later audio is hard to interpret.

Slide 2: Lab 1

- Explain that Lab 1 made the repo technically trustworthy by learning `z_content`, `z_style`, and a music gate.
- Phrase it simply: content is what the song is doing, style is how it sounds.
- Walk through the encoder split diagram while explaining the three outputs.
- Style probe 0.9417 means style is really concentrated in the style branch.
- Leakage 0.1083 means content is not secretly carrying style under another name.
- Gate ROC-AUC 0.9299 means the model learned musically valid structure well enough to trust later stages.
- End by saying that once style existed as a real signal, the next question was whether it could become a stable target.

Slide 3: Lab 2

- Explain that Lab 2 turned style from an embedding into a target space the generator could actually aim for.
- Point to the cluster figure first, then the centroid or target-vector diagram.
- Silhouette 0.4939 means the space is not one tangled cloud.
- Linear probe 0.8554 means the genre signal is accessible instead of buried.
- Nearest-centroid 0.8514 means the centroids behave like real genre anchors.

- End by saying that once representation and target space were in place, the remaining question became generation.

Slide 4: From Codec Editing to Diffusion Generation

- Introduce codec as the first strong proof that controlled reconstruction and editing were possible.
- Then explain that diffusion became the practical backbone because it gave the project a stronger realism anchor.
- Point to the side-by-side codec versus diffusion comparison.
- Play one short codec clip and one short diffusion clip.
- End by saying that once diffusion became the main path, the core tradeoff became obvious.

Slide 5: Short-Form Generation

- Use this slide to make the content-versus-style tradeoff feel obvious.
- Say clearly that style under-shoot is disappointing, but identity collapse is destructive.
- Point to the conservative / balanced / over-pushed comparison.
- Play a safe but style-light clip, then a balanced clip, then an over-pushed clip.
- End by saying that even promising local clips still had to survive time.

Slide 6: Long-Form Generation

- Explain that long-form accompaniment generation is the real remaining bottleneck.
- Say that local quality is not enough because seams, warble, drift, and instability build up over time.
- Point to the chunk timeline or seam diagram.
- Play one local-good clip and then the matching long-form failure.
- End by saying that this is why the best practical system came from engineering around the generator.

Slide 7: Hybrid Engineering

- Present the hybrid path as a systems solution, not a compromise in the bad sense.
- Explain that we preserved vocals where the model was fragile and generated accompaniment where style mattered most.
- Point to the preserved-vocal hybrid flow.
- Play the before / after practical comparison and then the best practical clip.
- End by saying that once we had a practical winner, the next question was whether later branches could beat it.

Slide 8: Later Experiments

- Frame this as the learning slide, not the graveyard slide.

- Explain that stronger style movement and new local textures were possible, but long-form stability stayed hard.
- Point to the compact family table and summarize what each knob improved or broke.
- Play only one or two representative later-branch clips.
- End by saying the conclusion is not that nothing worked, but that the hard part became clearer.

Slide 9: What We Actually Solved

- Keep this slide short and confident.
- Say that the project succeeded as a pipeline even though it did not fully solve long-form generation.
- State the four defensible wins: content/style representation, target-space control, locally plausible accompaniment generation, and a usable hybrid workflow.
- End by saying that this leaves one final question: what exactly still remains?

Slide 10: What Still Remains

- End on the real unresolved problem: long-form accompaniment generation.
- Say that realism, style separation, new instrumental character, and song identity still need to survive together over time.
- Do not play audio here; let the conclusion land cleanly.
- Final line: the next push should target long-form stability before even harder style pressure.